



Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management

Plan2Reality: Trusted Execution Intelligence for project controls

PROBLEM STATEMENT ID
SIH26122

THEME
Smart Automation

CATEGORY
Software

TEAM NAME
Team Seekers

TEAM LEADER
Daksh Verma

INSTITUTE
IIT Madras BS Degree Programme



Meet our team

Product, AI, backend, frontend, data and testing each have a named owner. Every member owns something visible in the final run-through.



Daksh Verma

Team Lead, AI and Product Engineer

3x Hackathon Winner, Product Engineer, Owns Product Planning and the L5/L6 matching engine



Preethy Parthasarathy

Domain and Evaluation

Wafer Assembly Director at Intel, 10+ years in process engineering, Owns ontology and benchmark.



Yash Jindal

Research & Presentation

Top 9, Asian Hack 2026. 3X Hackathon Winner. Owns the planner console, evidence view and review queue.



Vignesh Reddy Tadasina

Backend and Security

CTF placements at IIT Madras BS and IxEdge Forge. Owns ingestion, APIs, RBAC and the audit ledger.



Kartik Chilkoti

Full Stack and Testing

9x hackathon winner, startup mentor under DPIIT and Uttarakhand UCOST. Owns demo narrative and judge Q&A.



Chetna Nagar

Data and Gen AI

Web developer with internship experience, strong ML and GenAI depth, owns Data and Gen AI.



Novelty and Uniqueness



Our innovation is how the pieces work together.

- **Understand Field Language**

AI converts messy DPRs, text and voice into structured execution events.

- **Match with Schedule Context**

Identifiers + semantics + WBS + location + timing → exact

L5/L6 activity.

- **Trust Before Update**

Confidence + rule checks + evidence → Auto-post / Review / Unmatched.

- **From Update → Action**

Trusted progress → critical-path impact → delay forecast → recovery options.

One trusted loop:

Field Reality → Schedule Truth → Project Action



EXISTING SOLUTIONS

- **Primavera / project-control systems** - Strong at planning and schedule management, but rely on structured project updates.
- **Buildots / Doxel / OpenSpace** - Strong at visual site capture and progress measurement, but use a different, capture-heavy workflow.
- **Manual Excel / Email** - Flexible and universal, but slow, inconsistent and difficult to audit.

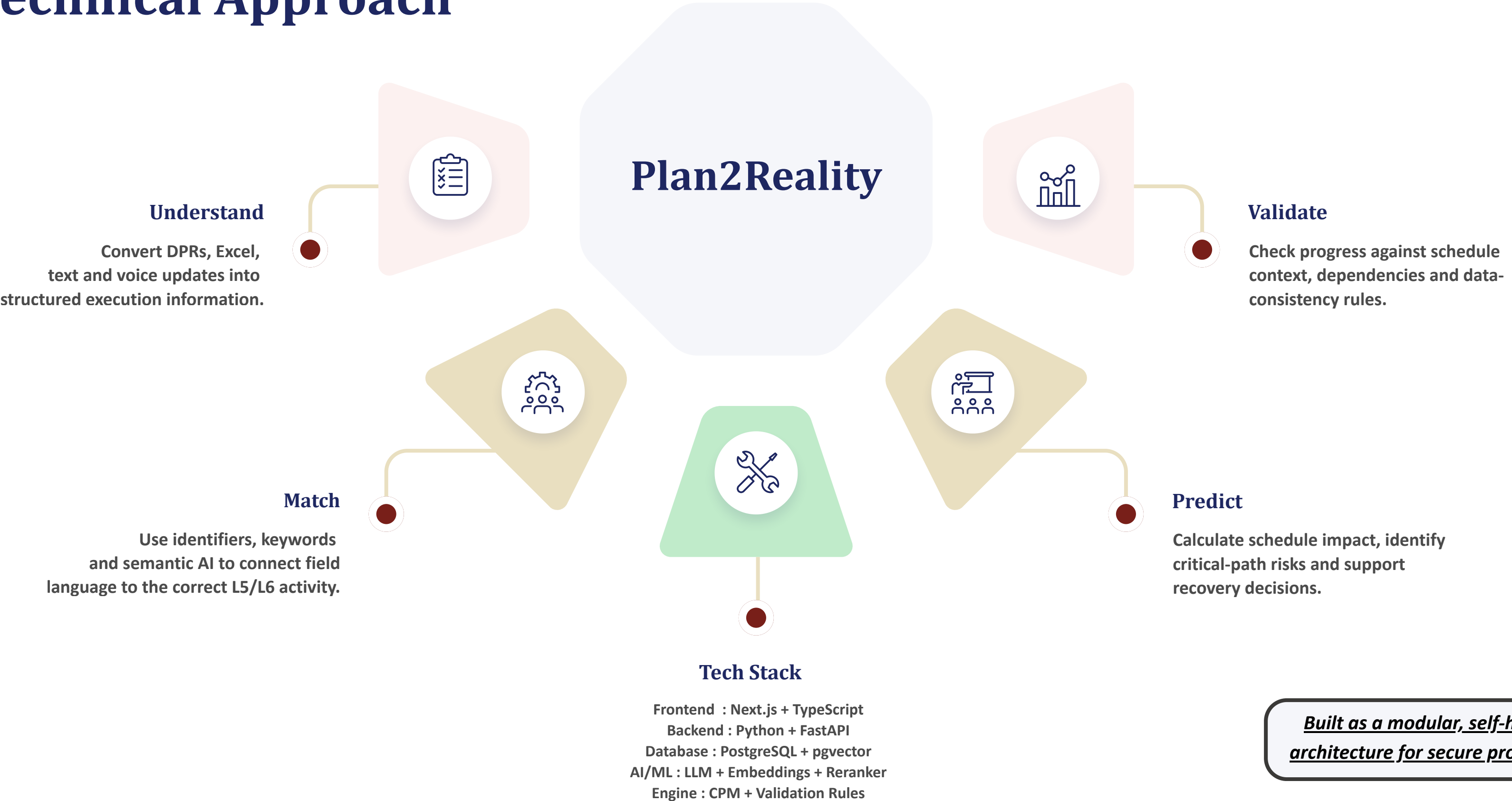


WHAT'S GENUINELY NEW

- Field language → exact schedule activity : Uses identifiers, semantics, location, discipline, WBS, timing & dependencies, not words alone.
- “I don’t know” is valid : NEW / UNMATCHED goes to planner review instead of forcing a bad match.
- Trust before update : Every update carries confidence, evidence, validation & approval history.
- From delay → decision : Shows what happens if we add manpower, change sequencing or add a shift.



Technical Approach

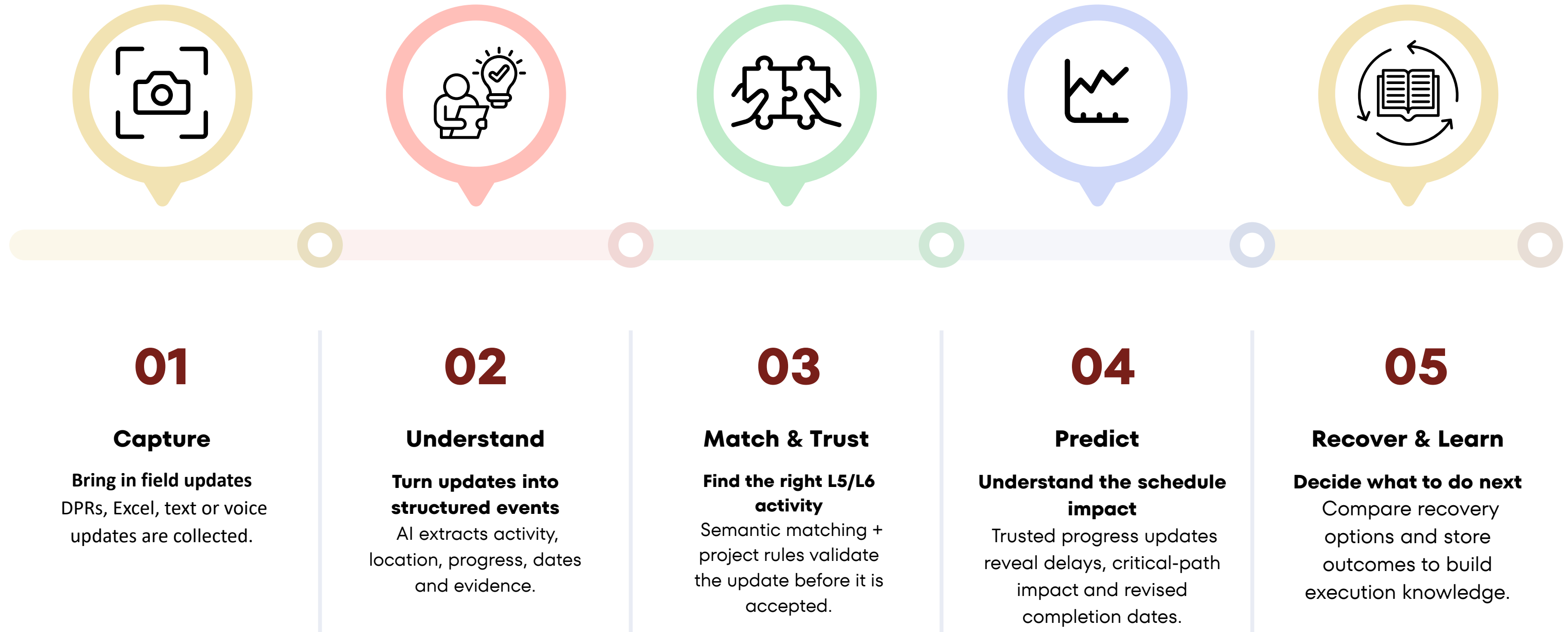


Built as a modular, self-hostable architecture for secure project data.



Process Flow

Five phases process



CAPTURE → UNDERSTAND → MATCH & TRUST → PREDICT → RECOVER & LEARN



Feasibility and Viability

i WHY IT IS BUILDABLE

- We are building the **hard part** first.
- The biggest technical risk is **reliable L5/L6 matching**, especially when reports are *incomplete, inconsistent or use different terminology*.
- *MVP SCOPE : 2-3 input formats + one complete project scenario.*

“We are not trying to build everything. We are concentrating our engineering effort on the one problem that makes the rest of the system valuable.”

o



Potential Challenges

- DIFFERENT WORDS, SAME WORK
Example:
“Spool erected”
vs.
“Line 24 erection”
 - FIELD DATA CAN BE WRONG :
Example:
Yesterday: 60%
Today: 45%
 - NO LIVE OIL DATA
- The PS does not provide live project data for the prototype.



Mitigation Strategy

- Hybrid matching using: identifiers + semantic similarity + schedule context
- Detect contradictions instead of blindly overwriting.
Confidence + rule checks + human review.
- Build a realistic benchmark with:
terminology variation + typos + missing IDs + conflicting progress + unmatched activities + date inconsistencies



Impact and Benefits

i

WHAT CHANGES FOR EACH PERSON?

- **Field Supervisor**

Reports naturally through voice, text or existing DPRs → no rigid new forms.

- **Planner**

Moves from manual reconciliation to reviewing only uncertain updates.

- **Project Manager**

Gets earlier visibility into delays, critical-path impact and recovery options.

- **Management**

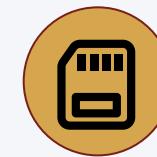
Gets a trusted, traceable view of project execution instead of scattered reports.

Less time reconciling. More time acting.



WHAT WE WILL MEASURE

- Activity matching accuracy
- Extraction accuracy
- Conflict detection
- Unmatched detection
- Schedule-impact accuracy
- Processing time
- Manual reconciliation time



Organization: Execution Memory

Every completed activity becomes reusable knowledge

**duration + delay reason + productivity + recovery
action + outcome**



Strategic Impact

From Project Tracking → Project Learning

Plan2Reality stores:

Delay causes + durations + recovery outcomes

→ helping teams learn from past projects and avoid repeated delays.

Research and References

Built on project controls, research and industry evidence



i Industry / Government

- Oil India Limited

Project-control and progress-reporting documentation

- Oracle

P6 EPPM REST / Web Services / XER integration documentation

- AACE International

Recommended Practice 37R-06 - Schedule Levels of Detail

i Industry Benchmarks

- Buildots, Doxel, OpenSpace, Primavera / Oracle, Procore
- Reviewed to understand existing project-progress and schedule-management approaches.

i Academic Research

- Amer, Jung & Golparvar-Fard (2021).
 - Transformer-based alignment of long-term and short-term construction plans.
- Pal et al. (2026).
 - Automatic mapping of schedule activities and reality models for construction progress tracking.

i Our Research Conclusion

- The opportunity is not “AI for construction.”
- The opportunity is reliable reconciliation between messy field language and structured project schedules.

“We studied both the project-control workflow and the existing technology landscape before deciding what not to build.”



Conclusion

Problem : Field updates are fragmented and often do not map cleanly to L5/L6 schedule activities, creating manual reconciliation and unreliable downstream visibility.

Solution

Plan2Reality converts field updates into verified, schedule-linked project progress and shows the impact of delays.

USP

Evidence-backed L5/L6 matching using project context, confidence, validation, unmatched handling and recovery decision support.

Feasibility

A focused prototype can demonstrate the complete workflow using 2-3 input formats, hybrid matching and deterministic schedule rules.

Impact

Faster reconciliation, more trustworthy progress visibility, earlier schedule-impact detection and better recovery decisions.